

STAT 101 Formula Sheet

$$CV = \frac{S}{\bar{x}} (100)\%$$

$$\bar{x}_w = \frac{\sum wx}{\sum w}$$

$$i = \frac{p}{100} n$$

$$\mu = E(x) = \sum xP(x)$$

$$\sigma_x = \sqrt{\sum [x - E(x)]^2 P(x)}$$

$$P(x) = \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

$$\mu = np$$

$$\sigma = \sqrt{npq}$$

$$q = 1 - p$$

$$P(x) = \frac{(\lambda t)^x e^{-\lambda t}}{x!}$$

$$z = \frac{x - \mu}{\sigma}$$

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$z = \frac{\bar{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

$$\bar{x} \pm z \frac{\sigma}{\sqrt{n}}$$

$$e = z \frac{\sigma}{\sqrt{n}}$$

Confidence Level	Critical Value
80%	Z = 1.28
90%	Z = 1.645
95%	Z = 1.96
99%	Z = 2.575